```php
<?php
    # Cucchi Francesco 5^AI index.php

    header("Location: login.php")
?>
```

```php
<?php
    # Cucchi Francesco 5^AI login.php

    if(!isset($_SESSION['role'])){
?>

<!DOCTYPE html>
<html>
    <head>
        <title>Esercizio 16</title>
    </head>

    <body>
        <h1>Pagina di Log-In</h1>
        <form method="POST" action="checkUser.php">
            Username: <input type="text" name="user" placeholder="User..." required>
            <br>
            Password: <input type="password" name="pwd" placeholder="Pwd..." required>
            <br>
            <input type="submit" name="login" value="ACCEDI">
            <input type="submit" name="reg" value="REGISTRATI">
        </form>

        <?php
        if(isset($_GET['errLog']) && $_GET['errLog'])
            echo "<p><b style=\"color:red;\"> Credenziali errate, ritenta l'accesso</b></p>";
        if(isset($_GET['errReg']) && $_GET['errReg'])
            echo "<p><b style=\"color:red;\"> Utente gi&agrave; esistente, impossibile completare la registrazione</b></p>";
        if(isset($_GET['regCompl']) && $_GET['regCompl'])
            echo "<p><b style=\"color:green;\"> Utente creato, procedere con il login</b></p>";
    }
            else{
        ?>
        <p><b>HAI GI&Agrave; EFFETTUATO IL LOGIN</b></p>
        <?php
                echo "<a href=\"home" . $_SESSION['role'] . ".php\"><button>TORNA ALLA HOME</button></a>";
            }
        ?>

        <br>
        <br>
        <br>
        <h4>-- DEBUG --</h4>
        <?php
            require_once 'func.php';
            print_r(importUsers());
        ?>
    </body>
</html>
```

```php
<?php
    # Cucchi Francesco 5^AI checkUser.php

    require_once 'func.php';

    if($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST) && !empty($_POST)){
        if(isset($_POST['login'])){
            $myUser = ["user" => $_POST['user'],
                       "pwd" => $_POST['pwd'],
                       "isAdmin" => false];
            $aus = isValidUser($myUser);
            if($aus != null){
                $myUser['isAdmin'] = $aus['isAdmin'];
                unset($aus);

                session_start();

                $_SESSION['user'] = $myUser['user'];
                $_SESSION['role'] = ($myUser['isAdmin']) ? "Admin" : "Cli";

                header("Location: home" . $_SESSION['role'] . ".php");
            }
            else{
                header("Location: login.php?errLog=true");
            }
        }
        else{
            if(checkUsername($_POST['user']))
                header("Location: login.php?errReg=true");
            else{
                writeUser($_POST['user'], $_POST['pwd'], "false");
                header("Location: login.php?regCompl=true");
            }
        }
    }
    else
        header($_SERVER['SERVER_PROTOCOL'] . " 418 I'm a teapot");
?>
```

```php
<?php
    # Cucchi Francesco 5^AI func.php

    function importUsers(){
        $rows = file("users.csv", FILE_SKIP_EMPTY_LINES);
        array_shift($rows);     # skip heading row
        $users = array();
        foreach($rows as $row){
            $aus = explode(";", $row);
            $users[] = [
                "user" => $aus[0],
                "pwd" => $aus[1],
                "isAdmin" => ($aus[2] === "false")
            ];
        }

        return $users;
    }

    function isValidUser($user){
        $allUsers = importUsers();
        $found = false;

        for($k = 0; $k < count($allUsers) && !$found; $k++){
            $found = ($user['user'] === $allUsers[$k]['user'] && $user['pwd'] ===
            $allUsers[$k]['pwd']);
        }

        return ($found) ? $allUsers[$k] : null;
    }

    function checkUsername($name){
        $allUsers = importUsers();
        $found = false;

        for($k = 0; $k < count($allUsers) && !$found; $k++){
            $found = $name === $allUsers[$k]['user'];
        }

        return $found;
    }

    function writeUser($user, $pwd, $isAdmin){
        $file = fopen("users.csv", "a");

        if($file){
            fwrite($file, $user.";".$pwd.";".$isAdmin.";".PHP_EOL);
            fclose($file);
        }
    }
?>
```

```php
<?php
    # Cucchi Francesco 5^AI homeAdmin.php
    session_start();
?>

<!DOCTYPE html>
<html>
    <head>
        <title>HOME ADMIN</title>
    </head>

    <body>
        <h1>HOME ADMIN</h1>
        <?php
            echo "<p>Benvenuto, amministratore " . $_SESSION['user'] . "</p>";
        ?>
        <a href="logout.php"><button>LOGOUT</button></a>
    </body>
</html>
```

```php
<?php
    # Cucchi Francesco 5^AI homeCli.php
    session_start();
?>

<!DOCTYPE html>
<html>
    <head>
        <title>HOME CLIENTE</title>
    </head>

    <body>
        <h1>HOME CLIENTE</h1>
        <?php
            echo "<p>Benvenuto, cliente " . $_SESSION['user'] . "</p>";
        ?>
        <a href="logout.php"><button>LOGOUT</button></a>
    </body>
</html>
```

```php
<?php
    # Cucchi Francesco 5^AI logout.php

    session_start();
    session_destroy();
    header("Location: login.php");
?>
```

```
username;pwd;isAdmin;
luca92;Tg3mVx8p;false;
silviaB;wK5r9uYb;true;
andrea01;Qp2tZh7c;false;
vale77;Ms8wRk3j;true;
mario_dev;Fd9tVp6n;false;
ginaX;Jr4mDc2z;false;
teoGame;yB1nLp8s;true;
nikoBeta;Cr5qTz9y;false;
sofia_art;Wm6gFx4d;false;
aliCoder;Qs3hNp8w;true;
marco2000;Tp7uRw2c;false;
francyQ;Bb2gVt5z;true;
giuly;Lp9eSf3v;false;
darioG;Vx4nRq6h;false;
nina42;Gq8kTw1b;true;
enzoDev;Mr5uXc7d;false;
rita88;Hs6vNp3z;true;
pietroR;Zw4eKl9t;false;
vale_js;Nq2mGy8h;true;
tommy_k;Rp9dFx4a;false;

```